

ActInf Textbook Group ~ Cohort 1 ~ Meeting 7

(Chapter 3 pt. 2)

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hello everyone it is june 16th 2022 and it's meeting seven of the active textbook textbook group cohort one we're in our second discussion of chapter three today we'll be looking at the questions that haven't been addressed so the ones without thumbs and also it's not too late to add more questions or to upvote questions and then additionally since the math learning group has been quite active in providing synchronous affordances for conversation around some of the background on the math as well as some aspects of the equations that we'll go through today and also even to like applications of the math and developing notebooks and such we'll take a look at the math in our move through chapter 3. before we go to the questions does anyone have anything that they just want to like remark on chapter 3 ali and then anyone else yeah well last week i talked about how i changed my mind about these bottom up and top down characterizations of low road and high road respectively after reading but it turns out that they are in fact bottom half and top down approaches after all you see i came across this uh in my opinion undeservedly lesser known piece of beautiful beautiful writing by carl fristan which explicitly argues for such characterizations and the philosophical importance of this distinction it's called beyond the desert landscape and it's inconspicuously hidden in this book andy clark and his critics so i i'd like to quote um a couple of sentences from this writing uh well he writes code the high road stands in for a top-down approach that starts by asking fundamental questions about the necessary properties things must possess if they exist using mathematical variational principles one can then show that existence is an embodied exchange of a creature with its environment that necessarily entails predictive processing as one aspect of the self-evidencing mechanics the low road is the purpose the agenda established by is the is to pursue the agenda established by content hellholtz and generalize in a bottom-up way the capacity for inference and prediction to see how far it takes us in understanding embodied exchange with the environment so if anyone likes to gain a deeper understanding about these conceptual um characterizations i highly highly recommend this chapter of this book and and

quite unusually for a first in writing uh it also lacks any mathematical formalism it doesn't include a single mathematical equation there thank you yeah in a recent stream we also talked about the desert landscape and clark's concern was that we might do away with what populates our implicitly assumed to be rich ecosystems of psychology philosophy and so on with important ecosystem players like goals and so the fear or concern was that we might be in a quiet and desert landscape if we replace some of these again implicitly or explicitly stated to be rich constructs like goals and we're like well it's just distribution matching then we've made some sort of like a desert monoculture and that would be um dip operate of what what what does clark say to his desert living critics um all right if anyone else has a thought they can raise their hand otherwise we can look at the questions okay so this first question if one defines preferred states as expected states one can say living organisms must minimize the surprise of their sensory observations i'm not clear on the ontology here is it that preference is the same as expectation or that formally we can use them in the same equation is that an ontological claim and is the free energy in physics an analogy or an ontological assertion about how things really are so one starting place was uh i found the slide and so wanted to highlight it this was in free energy a user's guide which was live stream number 37 um the slides will be probably a little bigger um it's worth restating just how unusual it is to interpret p as a measure of both probabilities and preferences they both start with p though there's nothing wrong with treating a distribution as a measure of preferences because distributions can have different interpretations but what's unorthodox and in need of justification is giving the same mathematical term two different interpretations within the same equation they then add some more details that unpack a little bit about how it is the case that the same term p can be utilized uh as uh both a probability and a preference and then they even extended it into a third p the p_h fitness where by allowing your probabilities and your preferences to converge to reduce the divergence between those two distributions then fitness is achieved and so they provided a restatement of the way that that is implemented and then uh highlighted an additional grounding or interpretive line so that's at least one part of that question and also to look towards the equations it's something that we'll be able to have very clear answers for and thanks to the work of ali and blue and others who have been annotating equations so for example in equation 3.1 or we can look at like box one equation one which is like a markov blanket definition in the description fields we have a natural language description and then the terms that are used in the description are those that are referenced using variables so that will help us translate across natural language of different human languages the discrete and

continuous time formalisms which we're going to explore in this textbook also identical or isomorphic or related formalisms in other papers which sometimes are using the same notation letters but not always but we could have another column for first in 2019 and here's first in 2013 and then there's a conversation to be had about where they align or don't and that's the kind of annotation work that somebody can engage in from the publications and that also helps us connect it to the the programmatic implementations like in active block rinse or in other scripts so that even that we can promote norms and practices around variable naming in math and in programming but also where there is divergences or differences which are important for innovating and evolving the norms we can also map them pretty coherently so all the equations for three can be um red naturally so we discussed this a lot in the math group and and again a lot of people contributed to this with one of the goals or functions being that whenever we are going to bring up an equation like if there's a question about equation 3.1 we can just briefly read the current state of the description because whether you're super familiar with the letters and variables and how they're concatenated or whether it's just like a first pass on your seeing this equation everybody will benefit when we align our regime of attention to actually what it says and then that will help us do many many other things but to return back to this does anyone want to raise their hand and give a thought on the relationship between preference and expectation ontological claims about preference and its relationship with expectation and more generally is the free energy something that isn't analogy or mathematical isomorphism or an equation that rhymes or is it also an assertion it's an equation that rhymes what makes you say that no not just because just because it's cute that's all uh i just joined the conversation but preference and expectation i wouldn't think they were at all referring to the same thing i mean expectation is a mathematical operator preference is a statement yes expectation is used in multiple ways it can refer to predictions about the mean or the mode or other attributes of a future time point like i'm expecting uh you know what are you expecting to happen on next monday right right also expectation can refer and that's how it's used with the fancy e to refer to the expected value of a distribution which for a gaussian is the mode in the center of the symmetric peak and that doesn't need to be about a future time point that can just be the expectation [Music] of the height of the classroom right now does it's not saying that's what we're expecting it to be in a year as it might be used conversationally but it's referring to just like the average of a distribution right mathematically it's an operator and verbally it can be a prediction or a desire or something and then yeah again in 37 and elsewhere it'll be worth returning to this again and again because it's one of the essential

architectural decisions in active inference that enable part of the necessity and sufficiency complex i don't know whether this is one one the whole um meat of it but it's part of what allows the dispensal of like an explicit reward term a homeostatic reinforcement reward driven learner would be like i'm more rewarded by being in my homeostatic range and then there needs to be this architecture around well how are we going to pursue reward alternatively an organism trying to maintain homeostasis realism or us modeling an organism that is maintaining homeostasis instrumentalism we can just say it expects to be in the range that conversationally we're going to describe as a preference how do we know we put the animal in the room and it's spending a lot of its time on this warmer side it prefers to be there we also expect it to be there we're less surprised when we see it so then it's possible to as they write here interpret that p location distribution in the room for the animal that we're observing or the temperature distribution for the body in the homeostatic context as being a probability density over temperatures over locations as well as reflecting the preferences not the qualia or the phenomenology or anything like too far out about what it means to prefer where preferences arise from whether they're fixed or learned in the model all these like many many trails that can come out of that but just at this like essential layer we don't need to interpolate a reward between what is expected and what is preferred any other comments from someone yes eric and then anyone else yeah i just um just to remark on that it seems you're still not removing the free variable of what the preference of the agent organism is you're just holding it you're putting in a different place so that you can use it in this unified equation but you still have this it's a pre variable okay what what do you want to do okay i'm going to call that an expectation and that's what i'm going to try to drive for when i uh optimize my my variational but it in that sense it's um it's kind of a convenience or it's a trick it's a little bit of an unnatural act i don't know like that way i think of it for a purpose there's nothing you haven't solved any other problem and you know the deeper problem of what should the organism attempt to achieve the goals thanks ali i think it also might be quite helpful to refer back to chapter 2 uh on page 36 i think yeah page 36 uh i think well here they have actually distinguished between the term prediction and exactly yes here the term prediction and expectation quite explicitly in terms of equation 2.6 so so let's look at equation 2.6 okay a few of the were all um that we don't have the third line fully described but just here's the third line what does this speak to with respect to preferences and expectations but here the preference actually is in the second term the callback library divergence here yeah it's actually inherent in the second term and the expectation expected ambiguity uh as it speaks for itself is the first term so we have here the

risk is defined as the divergence between the pre the preference and the states so i think that we can see the preference in this equation as well one note here is like that this g is equal to the first the seconds and um this line but one way that um we can understand the relationship of the second and the third line is like they're very similar the first term is identical here the only difference is that instead of a y we see an x and so this is like x being the hidden state and y being the observation in the case where it's a fully observable markov process then the outcomes and the hidden states are going to be like equal uncertainty about them in that case it's functionally that the hidden states are observables however if there's any amount of ambiguity like the a matrix between the hidden state and the observable then there's always going to be more uncertainty about the hidden state relative to the observable importantly on both sides there's a conditioning vertical line on policy selection and so that is what prevents us from getting into like well what if it's 95 correct or 96 or 97 correct or any of these sort of like model diminishing returns debates it's a functional that's calculated for each policy that have been been um possible of composing given the affordances a policy is just a sequence of actions so if there's four movement affordances there's four policies of time depth one there's 16 policies of time depth two and so on and for whatever in this discrete formalism finite set of policies there are they're evaluated here there's a policy selection that could choose the lowest expected free energy or it could proportionally choose from them with different kinds of waiting and so on but this is comparing the relative expected ambiguity and risk conditioned upon different specific policies and there's a process by which those specific policies can be enumerated and then this is a process by which they're going to be evaluated and then there's a policy selection by which they're selected organisms that fail to do this adequately enough will be dissipated or systems whether it's just um a molecular system or some other kind of statistical system that failed to do this well enough to remain being that kind of thing are not amenable to repeated measurements they are not that thing but things that remain being that thing for the time horizon of observations will be selecting policies that are consistent with this can ask a question about this yeah about this equation here say let the second line so i would have expected so first of all see the the preferences is not really spelled out very much in this chapter and i don't know where it will be but um i would have expected something more like preferences given observations like you know the observations of what's what's happening in the world that we can measure that's observation why and how does that and how does that fit with my preferences so what i want to optimize is my preference the achievement of my preferences as delivered by the observations now i understand in that that

in that second equation you're going to do a divergence between observations the uh the tildes the y tildes so you can't flip around p of c given y tilde there can somebody explain to me how this all fits together then how do you achieve preferences under that equation all give one um thought and and i absolutely agree to the first point which is like c is just i don't even know if it's described in this chapter like it just 34 i think 34. uh let me check it again i mean there is this yeah he says basically parameter includes preferences but that includes parameter yeah and also if you're uh if you go to uh section 7.4 i think yeah in that section it is untangled why these two parameters are related and how these two parameters are related to each other section yeah yeah i mean there's there's that's like multiple scales of where the information is not provided before when it needs to be provided which is like this i have made as just a helpful diagram of the partially observable markov decision process which is c is the preferences and it influences how g evaluates policies which are generated from affordances over a given time horizon policies intervene upon how hidden states change in the world and there's there's more in different readings but to return to um eric's question and then again anyone with the raised hand so the question was whether we're comparing outcomes or states how does this second term help us realize our preference well when is the kl we're trying to minimize g of π better policies have a lower score it's like um golf i guess or you could add negatives and make it higher um and also just it's an interesting note like climbing mount improbable biologists often think about the fitness peak and climbing to fitness peaks whereas for a physicist it's often framed and also like computer science gradient descending and local minima global minima but they're just actually like the same landscape with some negative signs so the kl divergence and here we're trying to minimize if it's zero that would mean like we are perfectly realizing our preference let's just say that there's two policies that we're comparing between and one of them is going to perfectly realize our preferences the other one isn't so we have like π one and π two and we're going to be calculating the kl divergence of the outcomes that we're expecting under the variational distribution queue under policy one and under policy two so if policy one is going to perfectly realize our our uh if the outcomes of policy one are expected to perfectly realize our preferences this term will be zero whereas if policy two is going to be expected to result in an outcome distribution that's not aligned with our preferences it doesn't get penalized here this is just evaluated as a as a value nor does it get penalized here because in this case of fixed preferences that's also not changing as we iterate through policies however there would be a divergence that would be non-zero like all divergences are between the inferior policy and the superior policy so by iterating over the

finite set of policies that we have taking this term and just leaving to the side for now only the kl divergence this will lead us to select policies that can be expected to bring alignment between the observations conditioned on policy and our preferences preferred outcome distribution thank you that's very helpful so just to kind of say that slightly differently is given your preferences that's that that's going to imply some outcomes and what you're trying to do with that kl that vergence is trying to get the policy that will achieve those outcomes if we expect and prefer to be within the homeostatic thermal range and then the two policies in the snowstorm are add the jacket or remove the jacket which one of those are expected to have the minimum divergence between us being in the homeostatic range and then the policy conditioned expected outcomes i'm going to have less divergence it doesn't even mean you're going to live it just means i'm going to have less divergence by having the jacket on in the snowstorm than by taking it off which is going to move the expected outcome distribution further from the preference distribution which here is fixed but doesn't have to be fixed so we have several um is anyone here the person who has written any of these questions so then we can just go to that one lyle there we go i think i'm i think i went in here okay so yeah i uh i wrote the one that is referring to the living organisms and the question about to what extent are these techniques valuable in what i would call social systems of wide-ranging sorts i know that this book at the outset limits itself to biological systems at least that's my recollection from the introduction but uh it just seems obvious to me that this would have much broader applicability i'm curious what sort of what the state of that is do i hesitate to put anyone on the spot john would you give a first response there while i add some resources my first thought on that is obviously our civilization is struggling and civilizations have failed in the past who collapsed so it's not necessarily the case that every social structure or societal structure is actually performing functional active inference but if we were a highly functional organism if societies were a highly functional organism we would be doing something like active inference and learning from learning from our mistakes and making sure that we put ourselves in a environment that uh that is sustainable and all that kind of stuff so that what my first thought is i think that all of this is applicable to societal systems but it doesn't necessarily mean that any existing societal systems actually perform functional active inference thanks it's almost like saying okay we're studying the penguins and their thermostats and homeostats this model could be parameterized to fit a thriving or a failing society along various different dimensions and so in the sense that this is like a linear aggression that can capture failing as well as succeeding societies yes the nested markov blanket formalism has qualitatively but

increasingly quantitatively been applied and there's multiple like startups and applications that are explicitly pursuing this angle it's been brought up for several years now and um in 2020 several of us wrote about it being applied to remote teams and organizations and communities i can add these links to decentralized science as a system and ecosystem so people have started to sketch that continent out and there's a lot more to understand what do you think lyle i think that's that makes a ton of sense i think uh obviously i guess the way i was thinking about the question is not so much whether to what extent social systems do this because social systems so you could take two points of view one is you could say social systems are behaving cognitively in the ways that we're discussing in other words they're they're our preferred states that are driving behavior now we could say those preferred states are the wrong states for whatever reasons so so there's there's a question you know between what we actually you know as in any particular social system was actually happening which i would argue could be described through this technique or what they should be doing which we could also describe through this technique right so i think i think it's important to have that distinction and there's a really long history of using dynamic systems models to explore those questions especially in the business context so that's mostly how i made my living through the through the decades though that that you know sort of you know as is to be or would prefer to be sort of questioned it just seems like this technique it would potentially be a significant advance over the tools in that space that currently exists which are primarily things like system dynamics and agent-based modeling stuff like that but this would seem to have the potential to look at things as they are why is it broken and what we might do yes so i've added several other uh references it's important and the book again is like if it's describing the linear model then there's the applications of a linear model and as people and as john brought up like many times in this uh series of live streams that we did together with four with just the two of us with john leading through basically the entire thread of this discussion of three papers and then two more group discussions there's been recent work on like scripts by mao and colleagues with a strong and a weak script formalization like a strong script being when the norms are practiced like in a ritualistic way like reading from a script versus the looser script concept which could include just norms and social praxis and then more recently their modeling work using pi mdp to talk about epistemic communities some of the very earliest live streams that we did were on narrative human communication cultural so these are absolutely topics that have been addressed although many of the earlier works are here's people in the world we're going to use active inference to model it but then there's so much more beyond that it's almost like a 1950s

cybernetics book well we can just imagine that the person is going to be involved in this feedback loop yeah but then that's not the the end of the discussion on that um ali and then john uh i just wanted to mention that uh well fep can potentially be applied to any dynamical system with ergodicity and uh because you see uh in any system with the ergotic behavior have uh some points of attraction some attractors that all the other states are converging to it's pretty general and well almost uh all dynamical systems we know of on different time scales can be potentially modeled with fep and as its corollary with active inference thank you that's the key point right because that's that's what we look at there got to your god ethnicity tough word for me um and and you can actually look at that even in the context of businesses business strategy and so forth with uh some understanding of the absorbing barriers right when when does a business dissipate when it's not close enough to its preferred state like adequate cash flow so um yeah so that i think that that just seems like a really interesting progression from where the state of the art of dynamic non-linear dynamic systems is the way we model businesses now so well that's the way some of us do it a lot of people use a spreadsheet but this would just seem like a potentially tremendous advantage yeah and then just a few notes then john um this stephen fox who's also joined on and talked about industrial engineering he's recently had this paper come out just a couple days ago and ironically a spreadsheet is provided in this paper but this is also in practice in several places in the world in startup ecosystems and it's very active influenced and then one other more conceptual note um cad cad which is the complex systems modeling software and i know some of you are involved in cad cad learning as well they describe it as a language for encoding generalized dynamical systems and that is what it is active block friends is our project where jacob has done much of the initial work that is implementing active inference in cad cad and we have a whole litany of questions about the relationships between base graphs generalized dynamical systems variational inference cad cad which takes more of like a sampling based approach versus the variational approaches that we're discussing here in active it's going to be very interesting as systems engineering complex systems simulation as well as tokenomic and other things becomes integrated with this analytical framework especially if we can demonstrate a true concordance between active inference and gds uh lyle i just wanted to say your your comments are really spot on earlier comments uh you had i think if i can paraphrase you had essentially said maybe even a dysfunctional society is performing active inference and then and then that would lead us to really to the the key components then of that society or that social system is what is its set of preferred states and where is it placing its attention so if the if

the if a uh say an elite group of very wealthy people or powerful people are running the world and they're running it for their preferences which is to maintain their wealth and to and to ensure that that they stay on top well then the world they could be performing functional active inference and the world could you know could uh could uh you know suffer the the problems that that kind of system would offer on the other hand if in a highly democratic system if people understand that they want clean air and clean water and peace and all that kind of stuff and then that's their set of preferred states and their attention is on that and they're performing active influence in thank you john um lyle and then mike i would just throw uh so john i'm completely on board with that i would i would just amplify a little bit on that i would say to a certain extent preferences and societal decisions are emergent so you you aren't going to find too many people in the world are going to say they want to be you know starving or have war whatever right and yet we do the the interesting thing is how to me is how does the individual you know i mean this is the the whole build up right how does the individual behavior somehow you create an emergent system that has many bad outcomes even though potentially none of those are bad actors some of them are bad actors but let's just take the the simple case there are no bad actors yet we get bad outcomes why is that and we can observe that in existing systems and then we can ask the question well you know why is that and then maybe if we understand why that is maybe we can do something to improve yes great um point and then just a short note there was in our remote teams we framed communication and information architectures as organizational affordances so that could assist with decision making find alignment where people do have similar preferences like a zone of proximate agreement in a negotiation context it might be broader but no matter how broad it is if you can't forage the path to it it's going to be a failing system mike yeah i just want to add i think scaling or the scale factor is hugely important here so uh if we're talking about interactions amongst a handful of individuals we're going to see different behaviors and outcomes than we might see on a global scale of cultural interaction okay and then the last comment on that john deal for it if your hand's still up i don't hear you actually okay your microphone may have changed or can other people hear john no okay um eric go for it yeah how does this all apply to a highly distributed system i don't see that like a society is good question the multi-agent case is something that hasn't been studied across the board but in 2021 with colleagues we made this is like one of the early it's not the first multi-agent simulation because there were multi-agent simulations however it was the first active inference model involving stigmergy which is the feedback process with agents modifying their environments and so on we can

think about even a nested distributed cognitive system as inheriting some sparsity from a causal perspective like there's more connections in the base graph within the nestmate's head as we're modeling it and then interactions induce like a sparsity between two brains and then they're always coupled with the environment and so on that sparsity is a statistical convenience but also it allows factorizability of models and the models may be factorized in a way free energy calculations could be for example summed within a level of collective behavior i don't know if that's going it's not that's not the solution in and of itself to the modeling but when we have systems of like nested markov blankets in closing systems and closing systems and then also lateral interactions and interfaces like collective behavior those systems are at least amenable to this kind of principled composition via their sparsity which does map onto a causal architecture or at least a useful one of the real world so also remains to be seen but i think that's a very important yeah i would throw it eric i would throw in on that that question because society of course broadly speaking is loosely coupled however of course there are clicks and densely coupled areas that are you know to create the well under well documented complex connection graphs right so that connection graph is emergent right so the question is i mean i think this would be actually a really fascinating research project can we show that fep would generate connection graphs or patterns the societal interactions similar to what has been observed right so there's lots and lots of research on what those graphs look like so it is it possible for us to or someone to use fep to show how those kind of connection patterns would emerge i think that's a really interesting question yes how we threshold parameterize those causal graphs how we evaluate counterfactuals on those graphs and that's related to the problem of structure learning but bayesian model selection and other sort of hierarchical bayesian methods have the tools there also just quick pointer to the project ideas table there's some very interesting projects that people have suggested jakub no big deal sounds like a good one but we can absolutely do it and that's one of the advantages of being in like an active and activity driven participatory research and education space is we're hopefully going to be able to help each other identify those projects and where their hazard hasn't been overlap and then scaffold those projects support them and then provide them like a place to communicate about the project playing into the ontology active block for ins all these pieces that one researcher working by themselves would have not those affordances let's just try to quickly look at the last several questions must all preferred states and strategies be ultimately linked to survival in some way or in the case of some organisms can free energy be related to something other than survival just a quick thought on that in the context of repeated

measurements survival is the capacity for repeated free energies for domain specific models can be made that have nothing to do with survival and i don't think that there's any contradiction there i still don't hear you actually uh rohan and then anyone else who raises their hand yeah hi uh so i was actually wondering because there's no treatment of time in any of these uh methods right so i like is there any accommodation made so if you're sampling something let's say you sample at a minute frequency you're getting a certain set of events whereas if you're sampling at a much higher frequency you would get a very different much more noisy data set but that might be essential to capture certain things that might actually enable survival maybe there's certain events that act only at certain times so uh from the perspective of repeated measurement right so i think free energy principle as it relates to survival uh could it just be related to some sort creating an internal clock for repeated measurement great question yes so far it's been almost time independent as you suggest we're going to be dealing with two ways to deal with time in chapter four and beyond and that's like a discrete time formalization where future and past states are explicitly modeled partially observable markov decision process and then also a continuous time variance where future and past time points are not explicitly modeled however a taylor series approximation higher and higher derivatives of a function are utilized to glean an approximation to the future behavior of a function those are two different ways of dealing with time and then you talked about like sampling rate i think uh you'll see many connections and ways to apply that for example you could imagine like if the sensor is every seconds but you're able to think 10 times per seconds or what about when the sensor is every 10 seconds but you're only able to think over this time scale and it's those kinds of situations where actually the frequencies are happening differently that we also might be able to address yes rohan uh yeah yeah also like if there are multiple sensors some of them react quickly some of them react slowly or if there's just or if we look at active inference as some sort of model averaging phenomena right so you have a model that runs really fast that reacts really quickly and produces predictions really quickly and the other one acts more slowly would those be valid targets in in any sense yes i think it's a very um interesting question i'll just add it here just because it was where we were discussing it um like about different time scales and the way that cognitive clock rates for example and sensory clock rates and then the time scale events the statistical regularities in the world happen in model stream 4.1 with devry's and colleagues they talked about the message passing in a factor graph so any time that there's a base graph there's a way to accomplish variational inference on that base graph using message passing techniques that's what their

package is about forney lab and then what they hinted at towards the end is actually this reactive message passing development which uses reactive programming techniques so whereas in message passing at least in the forney factor graph formalization it's like you crank the model and a message is passed amongst every node or there's like a given execution order of messages that are passed for the entire graph however the reactive message passing paradigm may allow like uncoupling of the frequency of message so you might have one prior that gets one message per hour and then there's another distribution that gets one message per minute and another one is getting one per second and even that one could be doing model averaging of something much faster so there's there's many ways that i think your questions will be starting to be explored rohan and then john okay john if you're like can you hear me now oh good on the question of survival i think you would have to identify to answer it you'd have to identify first what is the organism or what is the agent that you're talking about because obviously as if it's an individual person a person might uh sacrifice their life for others so then you know survival is not the main necessarily the main uh driver but if it's a society you know that's a that any individual might not survive but society as a whole might i suspect in the bigger picture that it's not it's not really survival of a species it's probably the ability to um the propagation of information somehow that's what i would guess in the in the biggest picture that's probably what everything is doing is propagating information and mike levin has also explored like how changes in the scope of the self can lead to like an organism with reduced fitness as a function of a cancer with increased fitness and vision so to speak and so like just because there's an imperative or normative way that we model behavior doesn't mean that everything works or anything like that ali yes adding to what john said well yes again i believe here survival doesn't necessarily mean biological survival and actually it means it it has a much broader sense and uh it means the existence of things actually it's a metaphysical concept here because um as i said carl priston has defined the existence or the the the requisites of the prerequisites of existence as uh having two uh parameters or two features uh namely uh being an ergotic system and having markup blankets so any system uh which has these any system which has these two properties can basically be considered uh as an existence system or we can call it a surviving system here just to in our last minutes before we go to tools touch on some of these points can markup blanket be considered a decoupling mechanism so can anything mathematical be considered a mechanism people use terms in different ways so just to raise the point that like a linear aggression on height and weight isn't a mechanism of height and weight it's just a descriptor so markov blanket doesn't need to be any

kind of mechanism at all it doesn't need to cut nature at the joints or anything like that that being said there are ways in which one could think of the blanket again variously depending on how the real generative model and generative process separated as part of a markov partition sometimes there's ways to think of it as how they're coupled other times it is a statistical decoupling of a type it's a partial information encapsulation but it's not a total encapsulation and that way it is like an interface and yes thanks to whomever has added this quote this is one reason why the transcripts of all of the live streams and discussions augmented with the ontology will be so helpful and searchable because we could find sentences that have raised similar points and addressed them so good question and then just last ones maybe we can come back to this and then um this one is a nice next step question thinking about friction what is the analogy to friction in a cognitive system what is what is the stochastic fluctuation on our cognitive path of least action what is that in the real world realism from the behavioral modeling perspective where are we using these equations partitioning variants into a path or a flow of least action and a variable that is analogous to stochastic fluctuation or friction but doesn't take on any of the baggage of describing some feature or component or mechanism of the physical system so also a great question though and we'll leave this one in case people want to continue to address it that brings us to the end of chapter three live meetings next week we'll head into two weeks of chapter four then two weeks of chapter five and then we'll have two weeks to sort of ground ourselves think about what we've learned talk about project ideas and then we'll be heading into the second cohort of the first half of the book and for those who are here who want to continue the the back nine as it were the second half of the textbook and also people are welcomed and encouraged to be facilitators or return participants in the second cohort first part because it's many coats of paint for everyone and so we want to provide that affordance to return again and again and again because the basics are the classics and the essence and the seed of innovation all this other important things that happen when we return and clarify also we've had to sort of do all of these entering the figures and equations for the first time but now that we have the architecture people can focus on annotating them or they can focus on adding new questions or improving the questions that others have written if anyone else has any comments that they want to make while the recording is still occurring i'll just give a few seconds for anyone to raise their hand um and then we'll end this session and then take a one minute break and then in this room we'll be continuing on with tools organizational unit and um if you want to continue speaking with people here about just any topic at all you can head up to a room up top and then of course

always you can join into the discord voice channels and just be working on things like during the math reading group or any other area rohan and then anyone else who raises their hand yeah so i actually so sorry for this late thought so i i'm just curious is so when you say the free energy principle it's basically figuring out some set point of states internal states that you that the agent wants to be with right it's not necessary that it actually couples to the real world so why wouldn't it be that okay it just becomes a brain in the vat type of situation so it does so it just find needs to find the set point that helps it survive it doesn't have to actually model the real world it just has to find a place no matter what happens just stay there so it's not really dynamic it is that could be is that a possibility it is dynamic it is explicitly modeling time and one can make a model just like a linear regression could be static or you could use the linear regression technique and use it in i was i was more referring it uh so uh if you look at some of the reinforcement deep reinforcement learning models right they usually find some trick within the game like if you're looking at atari games and you say maximize the score for example so it tries to find uh you know okay i can maximize the score by doing the simplest thing that doesn't require me to progress in any way it would increase the score so um yes these are important things so write them down so that we can have infinite discourse on them the brain the vat does still have an interface with the environment whether it's receive only and then you brought up like it doesn't have to be like modeling the quote real world this is the architecture that helps us bypass and say right the particulate states the particular states which are the blanket states and the internal states are doing inference acting as if they're doing inference on external states so it's perfectly compatible with scientific realism scientific surrealism aesthetic surrealism you could but so i mean the states could be hallucinating it could be hallucinating the world and it's actually just doing inference on the hallucination rather than actual reality as such anil seth and many others who speak of that uh reality as a controlled hallucination they're they're two feet in that territory without any qualms and feeling like it addresses questions that have been unaddressed by other perspectives okay okay yeah okay so i'll just put this down yep cool thank you everybody